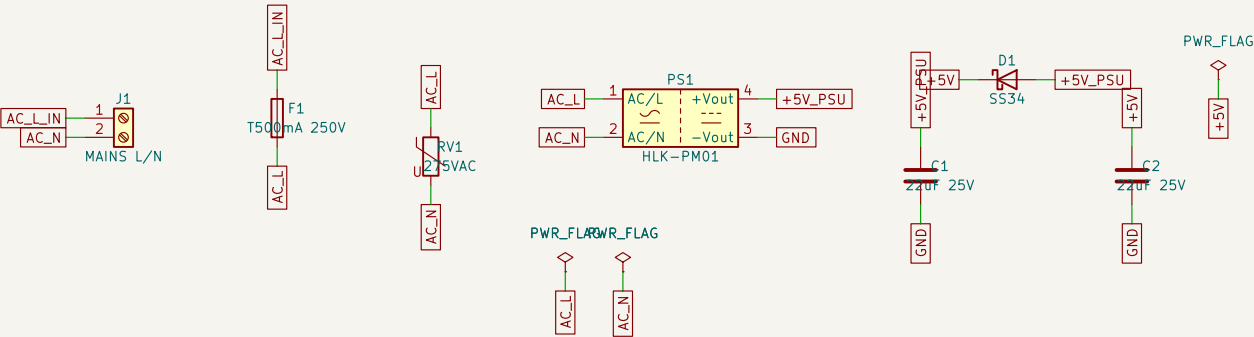
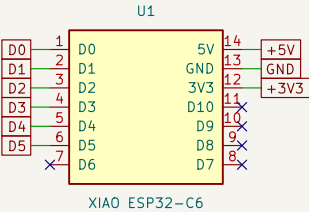


MAINS SIDE – 100–240 V AC. Keep ≥ 6 mm to everything on the low-voltage side.
F1 before everything; RV1 across L/N after the fuse, at the module input.

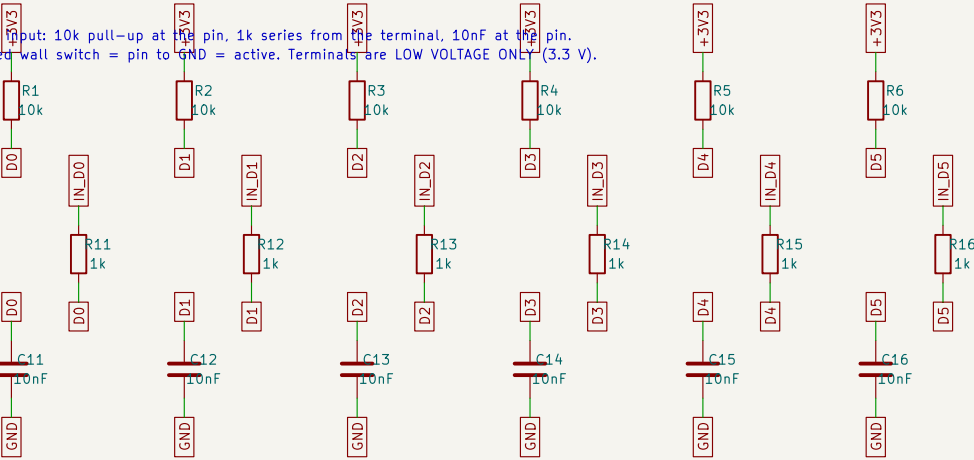
5 V from the certified, isolated HLK-PM01 (3 kV AC). D1 stops USB back-feeding it.



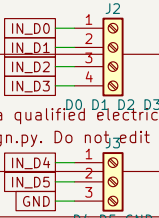
XIAO ESP32-C6. Pin map fixed by the firmware: D0–D5 = GPIO 0, 1, 2, 21, 22, 23.
D6–D10 unused. BOOT (GPIO9) and RESET are the buttons on the XIAO itself.



Each input: 10k pull-up at the pin, 1k series from the terminal, 10nF at the pin.
Closed wall switch = pin to GND = active. Terminals are LOW VOLTAGE ONLY (3.3 V).



NEVER connect USB while the board is on mains. Uncertified design: installation by an electrician.



Uncertified, AI-assisted mains design. Installation by a qualified electrician.
Generated by hardware/scripts/build_sch.py from design.py. Do not edit by hand.

github.com/tineira/hue-simple-switch

Sheet: /
File: hue-simple-switch-mains.kicad_sch

Title: hue-simple-switch mains carrier

Size: A4 Date: 2026-09-28

KiCad E.D.A. 10.0.6

Rev: 1.0

Id: 1/1